▶ 6.3 EULER'S EQUATION OF MOTION

This is equation of motion in which the forces due to gravity and pressure are taken into consideration. This is derived by considering the motion of a fluid element along a stream-line as :

Consider a stream-line in which flow is taking place in s-direction as shown in Fig. 6.1. Consider a cylindrical element of cross-section dA and length ds. The forces acting on the cylindrical element are:

1. Pressure force pdA in the direction of flow.
2. Pressure force $\left(p + \frac{\partial p}{\partial s} ds\right) dA$ opposite to the direction of flow.
3. Weight of element $\rho g dA ds$.

Let θ is the angle between the direction of flow and the line of action of the weight of element.

The resultant force on the fluid element in the direction of s must be equal to the mass of fluid element × acceleration in the direction s.

$$\therefore \quad pdA - \left(p + \frac{\partial p}{\partial s} ds\right) dA - \rho g dA ds \cos\theta$$

$$= \rho dA ds \times a_s \qquad \ldots(6.2)$$

where a_s is the acceleration in the direction of s.

Now $$a_s = \frac{dv}{dt}, \text{ where } v \text{ is a function of } s \text{ and } t.$$

$$= \frac{\partial v}{\partial s}\frac{ds}{dt} + \frac{\partial v}{\partial t} = \frac{v\partial v}{\partial s} + \frac{\partial v}{\partial t} \quad \left\{\because \frac{ds}{dt} = v\right\}$$

If the flow is steady, $\frac{\partial v}{\partial t} = 0$

$$\therefore \quad a_s = \frac{v\partial v}{\partial s}$$

Substituting the value of a_s in equation (6.2) and simplifying the equation, we get

$$-\frac{\partial p}{\partial s} dsdA - \rho g\, dAds \cos\theta = \rho dAds \times \frac{\partial v}{\partial s}$$

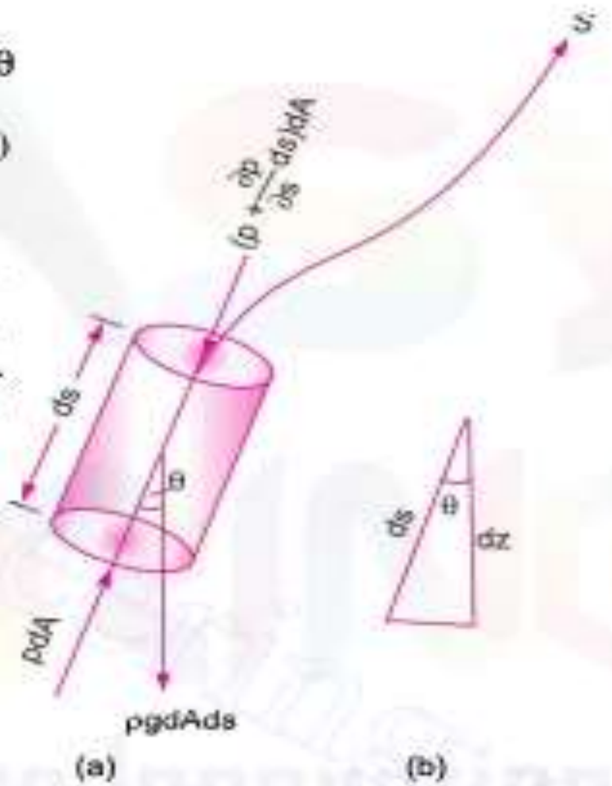


Fig. 6.1 *Forces on a fluid element.*

Dividing by $\rho ds dA$, $-\dfrac{\partial p}{\rho \partial s} - g\cos\theta = \dfrac{v\partial v}{\partial s}$

or
$$\frac{\partial p}{\rho \partial s} + g\cos\theta + v\frac{\partial v}{\partial s} = 0$$

But from Fig. 6.1 (*b*), we have $\cos\theta = \dfrac{dz}{ds}$

$$\therefore \qquad \frac{1}{\rho}\frac{dp}{ds} + g\frac{dz}{ds} + \frac{vdv}{ds} = 0 \quad \text{or} \quad \frac{dp}{\rho} + gdz + vdv = 0$$

or
$$\frac{dp}{\rho} + gdz + vdv = 0$$

Equation (6.3) is known as Euler's equation of motion.

▶ 6.4 BERNOULLI'S EQUATION FROM EULER'S EQUATION

Bernoulli's equation is obtained by integrating the Euler's equation of motion (6.3) as

$$\int \frac{dp}{\rho} + \int g dz + \int v dv = \text{constant}$$

If flow is incompressible, ρ is constant and

$$\therefore \qquad \frac{p}{\rho} + gz + \frac{v^2}{2} = \text{constant}$$

or

$$\frac{p}{\rho g} + z + \frac{v^2}{2g} = \text{constant}$$

or

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant} \qquad \text{...(6.4)}$$

Equation (6.4) is a Bernoulli's equation in which

$\frac{p}{\rho g}$ = pressure energy per unit weight of fluid or pressure head.

$v^2/2g$ = kinetic energy per unit weight or kinetic head.

z = potential energy per unit weight or potential head.

▶ 6.5 ASSUMPTIONS

The following are the assumptions made in the derivation of Bernoulli's equation :

(*i*) The fluid is ideal, *i.e.*, viscosity is zero (*ii*) The flow is steady

(*iii*) The flow is incompressible (*iv*) The flow is irrotational.

Problem 6.6 *The water is flowing through a taper pipe of length 100 m having diameters 600 mm at the upper end and 300 mm at the lower end, at the rate of 50 litres/s. The pipe has a slope of 1 in 30. Find the pressure at the lower end if the pressure at the higher level is 19.62 N/cm².*

Solution. Given :

Length of pipe, $L = 100$ m

Dia. at the upper end, $D_1 = 600 \text{ mm} = 0.6$ m

∴ Area, $A_1 = \dfrac{\pi}{4} D_1^2 = \dfrac{\pi}{4} \times (.6)^2$

$= 0.2827 \text{ m}^2$

p_1 = pressure at upper end

$= 19.62 \text{ N/cm}^2$

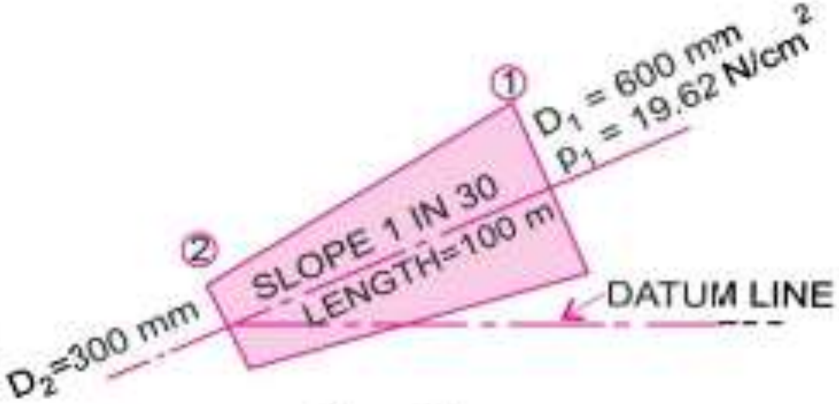


Fig. 6.5

$= 19.62 \times 10^4 \text{ N/m}^2$

Dia. at lower end, $D_2 = 300 \text{ mm} = 0.3 \text{ m}$

$\therefore$ Area, $A_2 = \frac{\pi}{4} D_2{}^2 = \frac{\pi}{4} (.3)^2 = 0.07068 \text{ m}$

$Q = \text{rate of flow} = 50 \text{ litres/s} = \frac{50}{1000} = 0.05 \text{ m}^3\text{/s}$

Let the datum line passes through the centre of the lower end.

Then $z_2 = 0$

As slope is 1 in 30 means $z_1 = \frac{1}{30} \times 100 = \frac{10}{3}$ m

Also we know $Q = A_1 V_1 = A_2 V_2$

$\therefore$ $V_1 = \frac{Q}{A} = \frac{0.05}{.2827} = 0.1768 \text{ m/sec} = 0.177 \text{ m/s}$

and $V_2 = \frac{Q}{A_2} = \frac{0.5}{.07068} = 0.7074 \text{ m/sec} = 0.707 \text{ m/s}$

Applying Bernoulli's equation at sections (1) and (2), we get

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

or $$\frac{19.62 \times 10^4}{1000 \times 9.81} + \frac{.177^2}{2 \times 9.81} + \frac{10}{3} = \frac{p_2}{\rho g} + \frac{.707^2}{2 \times 9.81} + 0$$

or $$20 + 0.001596 + 3.334 = \frac{p_2}{\rho g} + 0.0254$$

or $$23.335 - 0.0254 = \frac{p_2}{1000 \times 9.81}$$

or $p_2 = 23.3 \times 9810 \text{ N/m}^2 = 228573 \text{ N/m}^2 = \mathbf{22.857 \text{ N/cm}^2}$. **Ans.**

Problem 6.7 *A pipe of diameter 400 mm carries water at a velocity of 25 m/s. The pressures at the points A and B are given as 29.43 N/cm² and 22.563 N/cm² respectively while the datum head at A and B are 28 m and 30 m. Find the loss of head between A and B.*

Solution. Given :

Dia. of pipe, $D = 400 \text{ mm} = 0.4 \text{ m}$

Velocity, $V = 25 \text{ m/s}$

At point A, $p_A = 29.43 \text{ N/cm}^2 = 29.43 \times 10^4 \text{ N/m}^2$

$z_A = 28 \text{ m}$

$v_A = v = 25 \text{ m/s}$

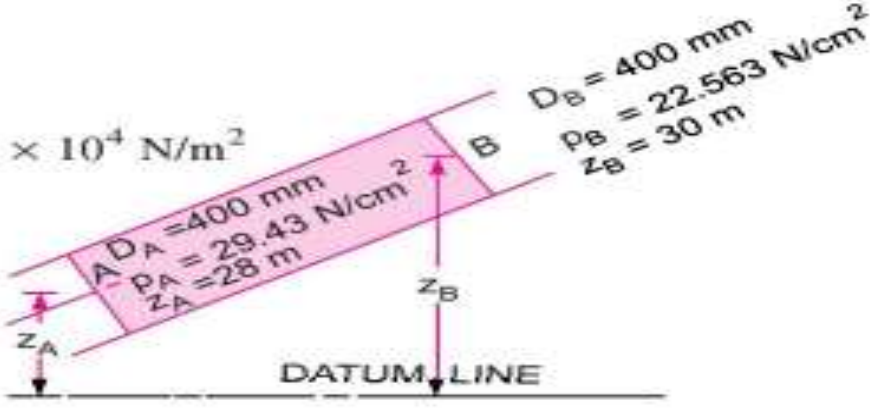


Fig. 6.6

∴ Total energy at A, $$E_A = \frac{p_A}{\rho g} + \frac{v_A^2}{2g} + z_A$$

$$= \frac{29.43 \times 10^4}{1000 \times 9.81} + \frac{25^2}{2 \times 9.81} + 28$$

$$= 30 + 31.85 + 28 = 89.85 \text{ m}$$

At point B, $p_B = 22.563 \text{ N/cm}^2 = 22.563 \times 10^4 \text{ N/m}^2$

$z_B = 30 \text{ m}$

$v_B = v = v_A = 25 \text{ m/s}$

∴ Total energy at B, $$E_B = \frac{p_B}{\rho g} + \frac{v_B^2}{2g} + z_B$$

$$= \frac{22.563 \times 10^4}{1000 \times 9.81} + \frac{25^2}{2 \times 9.81} + 30 = 23 + 31.85 + 30 = 84.85 \text{ m}$$

∴ Loss of energy $= E_A - E_B = 89.85 - 84.85 = $ **5.0 m. Ans.**

Velocity and Flow Measuring Devices

✓Bernoulli's Theorem

- This equation was developed first by Daniel Bernoulli in 1738.

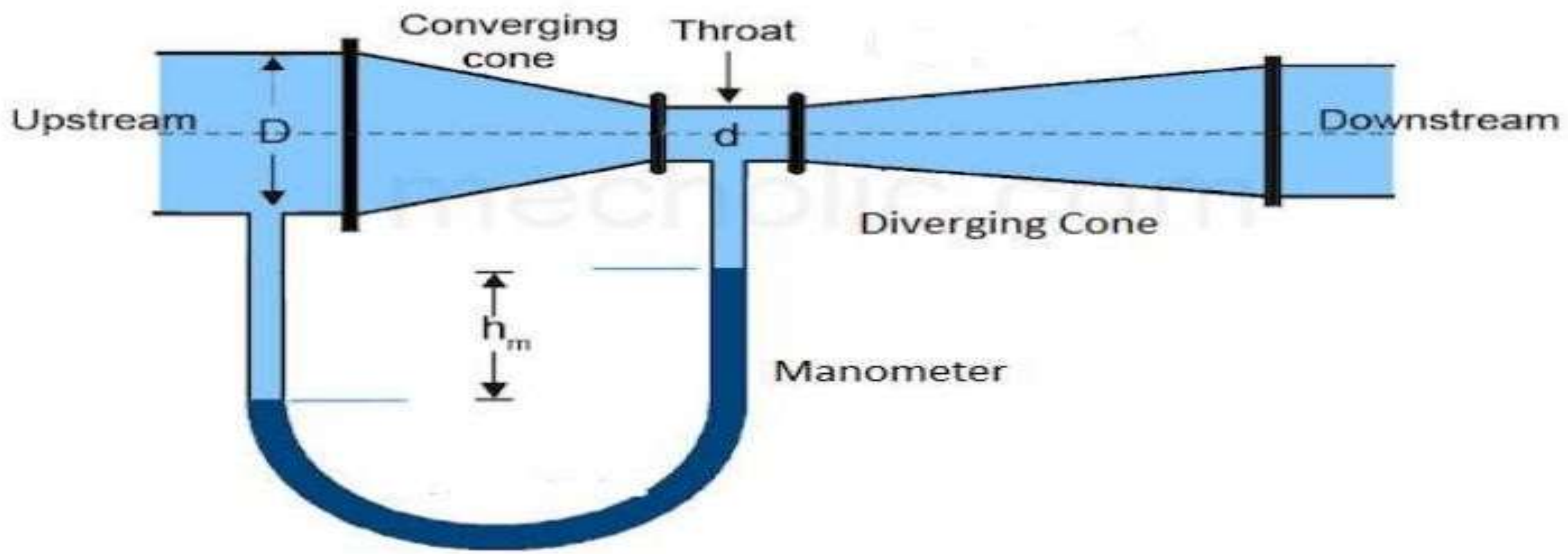
$$\frac{P}{\rho g}+\frac{V^2}{2g}+z=C_1(\text{constant})$$

Practical application of Bernoulli's law

- **Ventrurimeter**
- **Orificemeter**
- **Pitot tube**

VENTURIMETER

- Venturimeter is a device used for measuring the rate of flow (discharge) of fluid flowing through the pipelines.
- Venturimeter is work on Bernoulli's equation and its simple principle is when velocity increases pressure decreases.



<u>It consist of 3 parts:</u>

- a) Converging Cone.
- b) Short cylindrical throat.
- c) Diverging cone.

Converging Cone:

- The function of converging cone is to accelerate the flow and create pressure difference between the inlet to converging cone and the throat.
- Angle of Diverging cone is from 14° to 20°.

Short Cylindrical throat:

- The function of throat is to stabilize the flow
  and facilitate the provision of pressure tapping.
- Diameter Ratio:

$$d/D = 0.4 \text{ to } 0.7$$

Where, D= Diameter of Pipeline
d= Diameter of throat

Diverging cone:

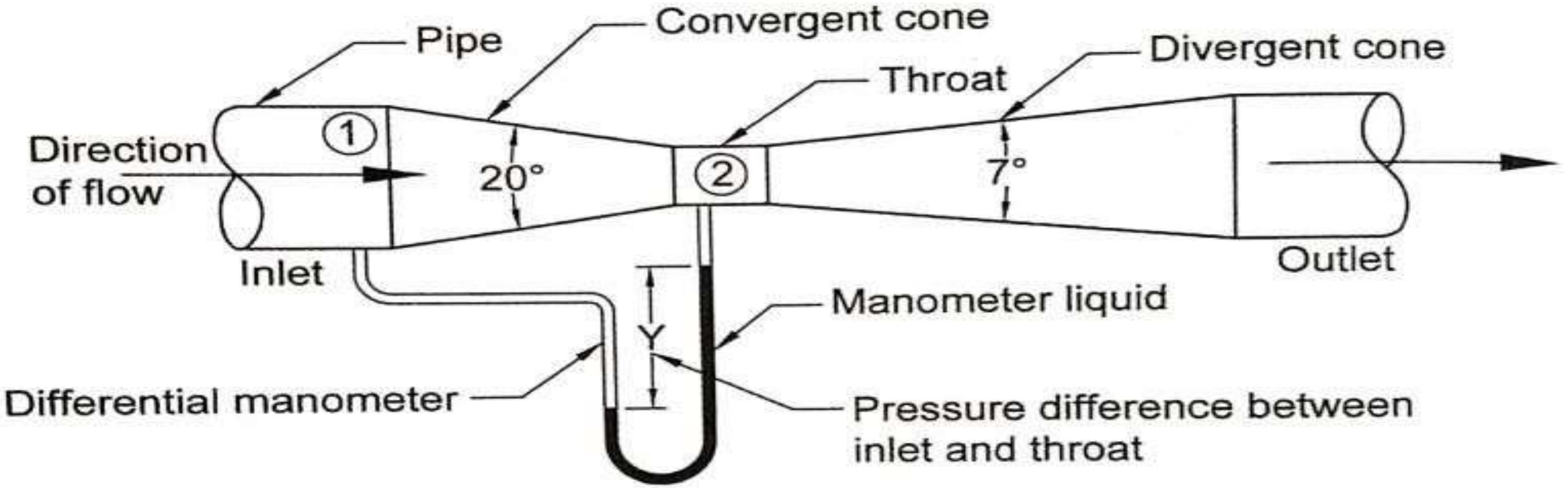
- The function of diverging cone is to reduce the velocity and increase the pressure to its original value to the extent it is possible practically.
- Angle of Diverging cone is from 5° to 7°

Working

- **Cross sectional area of throat section is smaller than inlet section due to this the velocity of flow at throat section is higher than velocity at inlet section.**
- **The increases in velocity at the throat result in decreases in pressure at this section , due to this pressure difference is developed between inlet and throat of the venturimeter.**
- The pressure difference measure between the entry section 1 and the throat section 2, usually by means of a U-tube manometer.
- The axis of venturimeter may be horizontal or inclined or vertical.
- Area of flow in divergent section is increased gradually to avoid separation of flow and reduce friction losses.

Expression for the rate of flow through venturimeter:-

- *Let D, p1, V1 & A1, are the diameter at the inlet, pressure at the inlet, velocity at the inlet and area at the cross section 1. And d, p2, V2 and A2 are the corresponding values at section 2.*



Applying Bernoulli's equation at inlet and throat of the venturimeter.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\left(\frac{P_1}{\rho g} + Z_1\right) - \left(\frac{P_2}{\rho g} + Z_2\right) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

Piezometer head difference, $h = \left(\frac{P_1}{\rho g} + Z_1\right) - \left(\frac{P_2}{\rho g} + Z_2\right)$

Therefore, $h = \frac{V_2^2}{2g} - \frac{V_1^2}{2g} = \frac{V_2^2 - V_1^2}{2g}$

$$V_2^2 - V_1^2 = 2gh \quad \text{.................................} \quad (1)$$

Applying continuity equation between inlet and throat of the venturimeter.

Discharge, $Q = A_1 V_1 = A_2 V_2$

$$V_1 = \left(\frac{A_2}{A_1}\right) V_2$$

Substituting the value of (V_1) in equation (1)

$$V_2^2 - \left(\frac{A_2}{A_1}\right)^2 V_2^2 = 2\,g\,h$$

$$V_2^2 \left(1 - \frac{A_2^2}{A_1^2}\right) = 2\,g\,h$$

$$V_2^2 \left(\frac{A_1^2 - A_2^2}{A_1^2}\right) = 2\,g\,h$$

$$V_2^2 = \frac{A_1^2 \times 2gh}{\left(A_1^2 - A_2^2\right)}$$

$$V_2 = \frac{A_1\sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

Then, Discharge, $Q = A_2 V_2$

Substituting the value of (V_2) in discharge equation, $Q = \frac{A_2 \times A_1\sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$

Since Bernoulli's equation have applied without considering losses, obtained discharge is called the theoretical discharge.

Theoretical discharge, $Q_{the} = \frac{A_1 A_2\sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$

Co-efficient of discharge, $= \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$

$$C_d = \frac{Q_{act}}{Q_{the}}$$

Actual discharge, $Q_{act} = C_d \times Q_{the}$

$$Q_{act} = \frac{C_d \times A_1 A_2\sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

$$\text{Co-efficient of discharge,} = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$$

$$C_d = \frac{Q_{act}}{Q_{the}}$$

$$\text{Actual discharge, } Q_{act} = C_d \times Q_{the}$$

$$Q_{act} = \frac{C_d \times A_1 A_2 \sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

Value of ‘h’ is given by differential manometer

- Case I: Let differential Manometer contains liquid which is heavier than liquid flowing through pipe

Sm= Specific gravity of manometer liquid , Sw =specific gravity of liquid flowing through pipe

Y=difference of heavier liquid in column

h=Y[Sm / Sw -1]

Case II :If differential Manometer contains liquid lighter than liquid flowing through pipe:

h=Y[1- Sm / Sw]

The value of coefficient of discharge varies from 0.95 to 0.99.

- Merits:
- Recovery of Pressure is near original value
- The discharge coefficient is high
- Loss of energy is minimum.
- It can be installed vertically, horizontally, inclined.
- They are more precise and can be used for a wide range of flows.
- Demerits:
- Initial costs, installation and expensive maintenance.
- Occupies more space than orifice meter

Example A horizontal venturimeter with inlet diameter 20 cm and throat diameter 10 cm is used to measure the flow of oil of specific gravity 0.8. The discharge of oil through the venturimeter is 60 litres/s. Find the reading of the oil-mercury differential manometer. Take co-efficient of venturimeter, $C_d = 0.98$.

Given data

Inlet diameter of venturimeter, $D = 20\text{ cm} = 0.20\text{ m}$

Throat diameter of venturimeter, $d = 10\text{ cm} = 0.10\text{ m}$

Specific gravity of oil, $S_w = 0.8$

Discharge of oil through the venturimeter, $Q_{act} = 60\text{ litre/s} = 0.060\text{ m}^3\text{/s}$

Co-efficient of discharge, $C_d = 0.98$

Reading of Oil-mercury differential manometer, $x = ?$

$$Q_{act} = \frac{C_d A_1 A_2 \sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}}$$

$A_1 = \frac{\pi}{4} D^2$
$= \frac{\pi}{4}(0.20)^2$
$= 0.0314\ m^2$

$A_2 = \frac{\pi}{4} d^2 = \frac{\pi}{4}(0.10)^2$
$= 0.00785\ m^2$

$$\therefore 0.060 = \frac{0.98 \times 0.0314 \times 0.00785 \sqrt{2gh}}{\sqrt{(0.0314)^2 - (0.00785)^2}}$$

$h = ?$

$h = 2.910\text{m of oil}$

Differential manometer eqn, $h = x\left(\frac{S_m}{S_w} - 1\right)$

where, $x = ?$
$S_m = 13.6$
$S_w = \text{sp. gr. of oil} = 0.8$

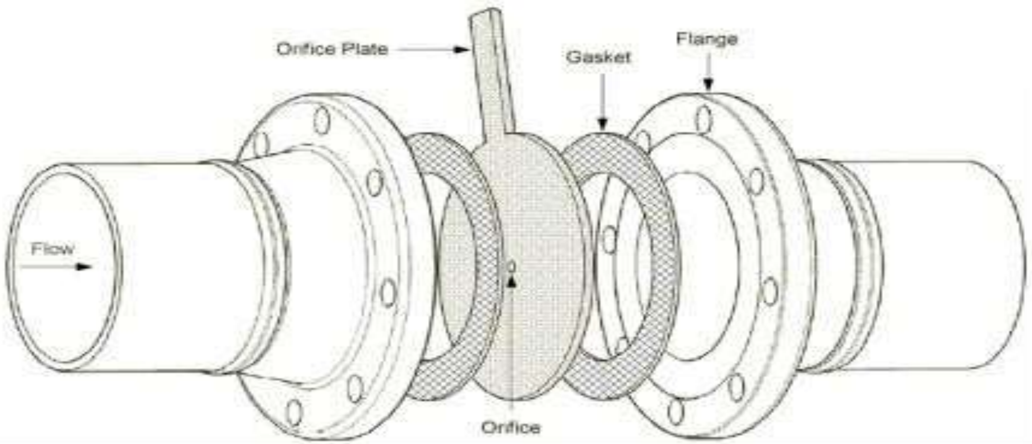
$\therefore h = x\left(\frac{S_m}{S_w} - 1\right)$

$2.910 = x\left(\frac{13.6}{0.8} - 1\right)$

$\therefore x = 0.1818$ metres of mercury
$= 18.18$ cm of mercury

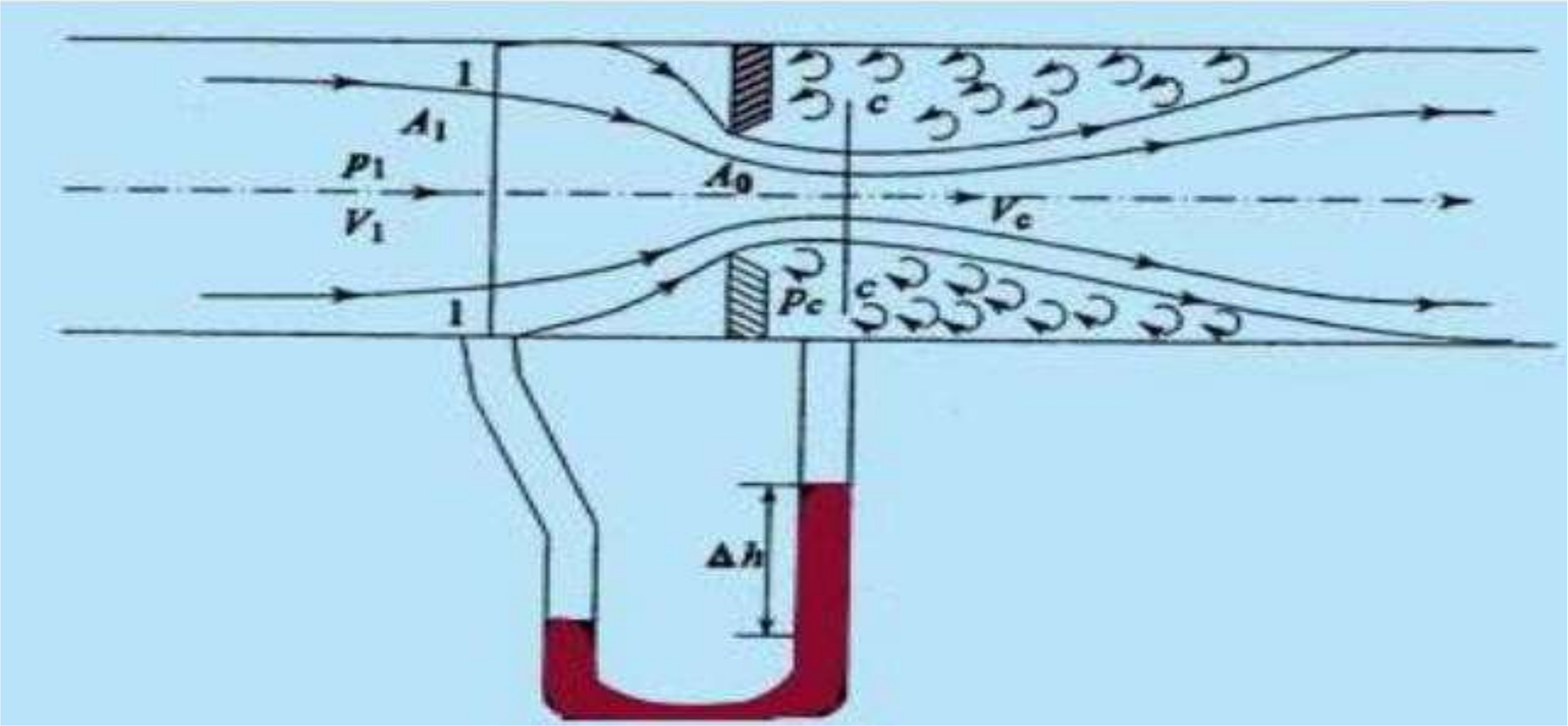
Orifice meter/Orifice Plate

- It is a device used for measuring rate of flow of fluid through a pipe.
- It is cheaper in comparison to venturimeter but it works on same principle as venturimeter.
- It consists of a flat circular plate which has a circular sharp edged hole called orifice which is concentric with the pipe.
- The plate is held in the pipeline between two flanges.The orifice diameter is generally kept half of the diameter of the pipe
- The value of Cd varies between 0.60 to 0.65.
- It is a economical and less space is required for fitting.



Orifice meter

- It may be installed in pipeline with a minimum of trouble and expense



Working:

- A pressure difference is created by reducing the cros-sectional area of the flow passage and the measurement of pressure difference enables the determination of discharge through pipe.
- The pressure difference is measured by using a U – tube differential manometer

COMPARISON BETWEEN VENTURIMETER AND ORIFICEMETER

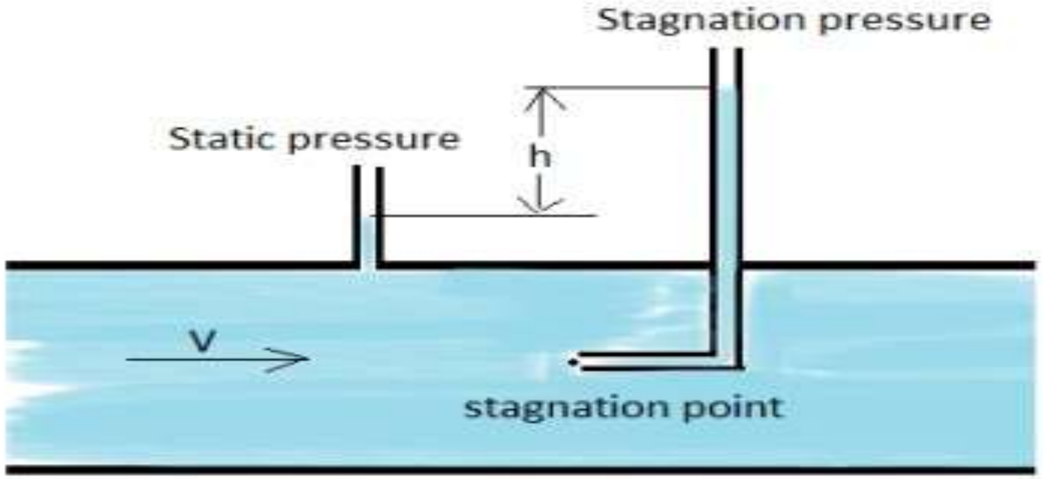
	VENTURIMETER	ORIFICE METER
1.	Used for measuring discharge of fluid flowing through a pipe line.	Used for measuring discharge of fluid flowing through a pipe line.
2.	Works on the principle that reducing the area of the flow passage (throat), a pressure difference is created.	Works on the principle that reducing the area of the flow passage (orifice), a pressure difference is created.
3.	It consists of a converging cone, throat and diverging cone.	It consists of an orifice plate and orifice.
4.	Gradual expansion of fluid takes place in the divergent cone, downstream section of the flow passage.	Sudden expansion of fluid takes place in the down stream section of the flow passage from orifice to the diameter of pipe.

Pitot tube

- It is a device used for measuring the velocity of flow at any point in a pipe/channel.
- It is based on the principle that

"When velocity of flow at a point becomes zero, the pressure there is increased due to conversion of kinetic energy into pressure energy"

- The common type of pitot tube consists of a glass tube bent at right angles.



The liquid rises up in it due to conversion of kinetic energy into pressure energy. The velocity is measured by measuring the rise of liquid in the tube.

Static pressure – Stagnation pressure and Dynamic pressure

- Static Pressure – It is the pressure of fluid which is the normal force exerted by the fluid on unit area of surface.

  - Static pressure is independent of the fluid movement or flow

- Dynamic Pressure – It is associated with the velocity of the flow or the flow of the fluid.

  - It is the amount of kinetic energy converted in to pressure energy when the fluid comes to rest.

- Stagnation Pressure – It is the total pressure of the fluid

  Stagnation pressure = Static pressure + Dynamic pressure

Expression for velocity of flow

Consider two points (1) and (2) at the same level in such a way that point (2) is just as the inlet of the pitot-tube and point (1) is far away from the tube.

Let p_1 = intensity of pressure at point (1)

v_1 = velocity of flow at (1)

p_2 = pressure at point (2)

v_2 = velocity at point (2), which is zero

H = depth of tube in the liquid

h = rise of liquid in the tube above the free surface.

Applying Bernoulli's equations at points (1) and (2), we get

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

But $z_1 = z_2$ as points (1) and (2) are on the same line and $v_2 = 0$.

$$\frac{p_1}{\rho g} = \text{pressure head at (1)} = H$$

$$\frac{p_2}{\rho g} = \text{pressure head at (2)} = (h + H)$$

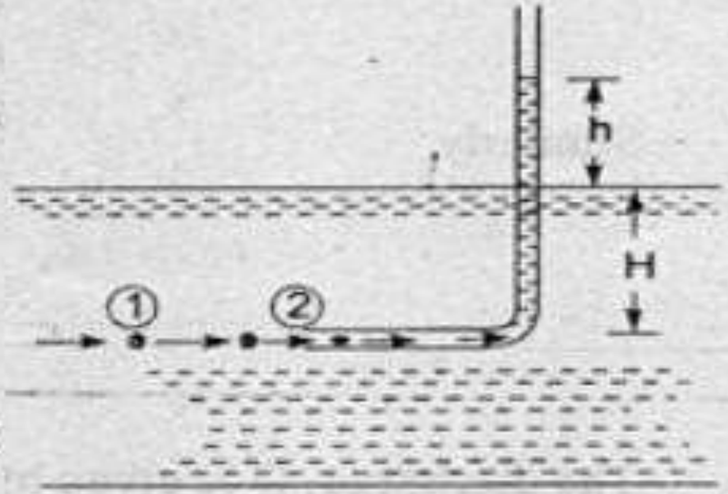


Fig. 6.13 *Pitot-tube.*

Substituting these values, we get

$$\therefore \quad H + \frac{v_1^2}{2g} = (h + H) \quad \therefore \quad h = \frac{v_1^2}{2g} \quad \text{or} \quad v_1 = \sqrt{2gh}$$

This is theoretical velocity. Actual velocity is given by

Expression for velocity of flow

$$(v_1)_{act} = C_v \sqrt{2gh}$$

where C_v = Co-efficient of pitot-tube

$\therefore$ Velocity at any point $\quad v = C_v \sqrt{2gh}$

A submarine moves horizontally in sea and has its axis below the water surface. A pitot tube is placed in front of the submarine along its axis is connected to the two limbs of a U-tube containing mercury. The difference in mercury level is found to be 170 mm. Find the speed of submarine in Km/hr, knowing that specific gravity of sea water is 1.025.

Given data:

Difference in mercury level in the U-tube, Y = 170 mm = 0.170 metres of mercury

Specific gravity of sea water = 1.025

Solution:

The equation of the pitot-tube is given by, $V = C_v \sqrt{2gh}$

Assume, co-efficient of velocity, $C_v = 1$

h = Dynamic pressure head in metres of sea water

Y = Difference in mercury level (dynamic pressure head) in metres of mercury.

The manometer equation is given by, $h = Y\left(\frac{S_m}{S_w} - 1\right)$

where, S_m = Specific gravity of mercury = 13.6

S_w = Specific gravity of sea-water = 1.025

$$h = 0.170\left(\frac{13.6}{1.025} - 1\right) = 2.085 \text{ metres of sea-water}$$

$$V = 1 \times \sqrt{2 \times 9.81 \times 2.085} = 6.395 \text{ m/s} = \frac{6.395}{1000} \times 3600 = 23 \text{ km/hr}$$

7.10 MOMENTUM EQUATION (IMPULSE MOMENTUM THEORY)

The momentum equation is another very useful equation which is used in fluid dynamics, in addition to the continuity equation and energy equation. The application of momentum equation leads to the solution of several fluid flow problems. In fluid dynamics, it may be required to determine force exerted on solid surface by the action of the flowing fluid. The magnitude of such force may be determined by momentum equation. The force exerted by a jet of water, striking on moving curved vanes of hydraulic machines like turbines and centrifugal pumps is an important example for the application of momentum equation.

The momentum equation is based on the law of conservation of momentum or momentum principle (Newton's second law of motion). *The momentum principle which states that the net force acting on a fluid mass is equal to the rate of change of momentum of fluid flow in the derection of force.*

We know that, momentum is the product of mass and velocity of the fluid.

Momentum = Mass × velocity = $m \times V$

The rate of change of momentum = $\frac{d}{dt}(m \times V)$

According to Newton's second law of motion (momentum principle)

The net force acting on the fluid = Rate of change of momentum

$$F = \frac{d}{dt}(m \times V)$$

For a constant fluid mass, m = constant

$$F = m.\left(\frac{dV}{dt}\right)$$

$$F = m.\ \frac{(\text{Final velocity of fluid} - \text{Initial velocity of fluid})}{\text{Time period}}$$

$$F = \frac{m}{t}\,(\text{Final velocity of fluid} - \text{Initial velocity of fluid})$$

where, $\frac{m}{t}$ = Mass flow rate = $\rho.Q$

ρ = Density of the fluid in kg/m³ $\quad$ Q = Discharge of fluid in m³/s

Momentum equation can be written as;

$$F = \rho\ .\ Q\,(V_2 - V_1)$$

where, V_2 = Final velocity of fluid $\quad$ V_1 = Initial velocity of fluid

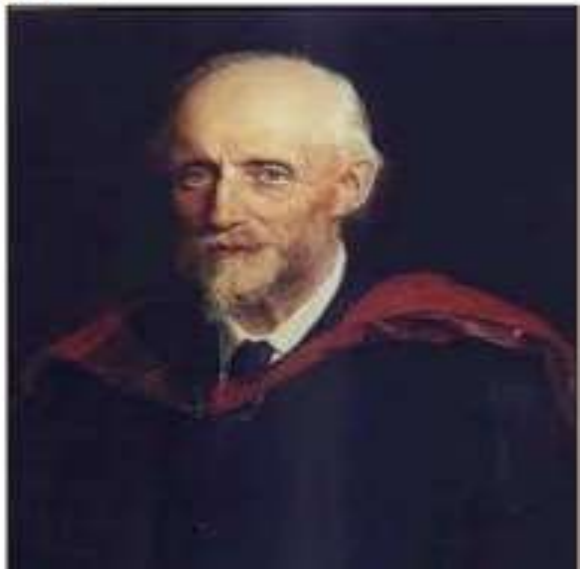
PIPE FLOW

- A pipe is used for carrying fluids under pressure
- It is a closed conduit of generally circular cross-section. It is able to withstand higher pressure differentials without distortion
- The flow through a pipe is called pipe flow when it runs full with the fluid
- Pipes are used in the field of industrial as well as domestic purposes

INTRODUCTION

- Prof. Osborne Reynolds conducted the experiment in the year 1883.
- This was conducted to demonstrate the existence of two types of flow :-

    1. Laminar Flow
    2. Turbulent Flow

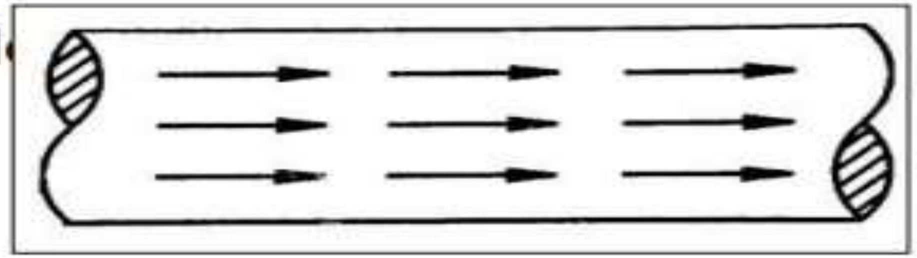


1. Laminar Flow:-

Laminar flow is defined as that type of flow in which the fluid particles move along well-defined paths or stream lines and all the stream lines are straight and parallel.

Factors responsible for laminar flow are:-

- High viscosity of fluid.
- Low velocity of fl
- Less flow area.

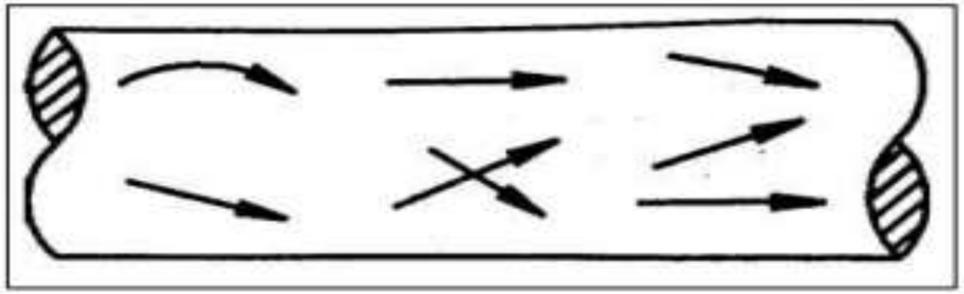


For example,

- Flow through pipe of uniform cross-section.

2. Turbulent Flow:-

Turbulent flow is defined as that type of flow in which the fluid particles move is a zigzag way. The Fluid particles crosses the paths of each other.



For example,

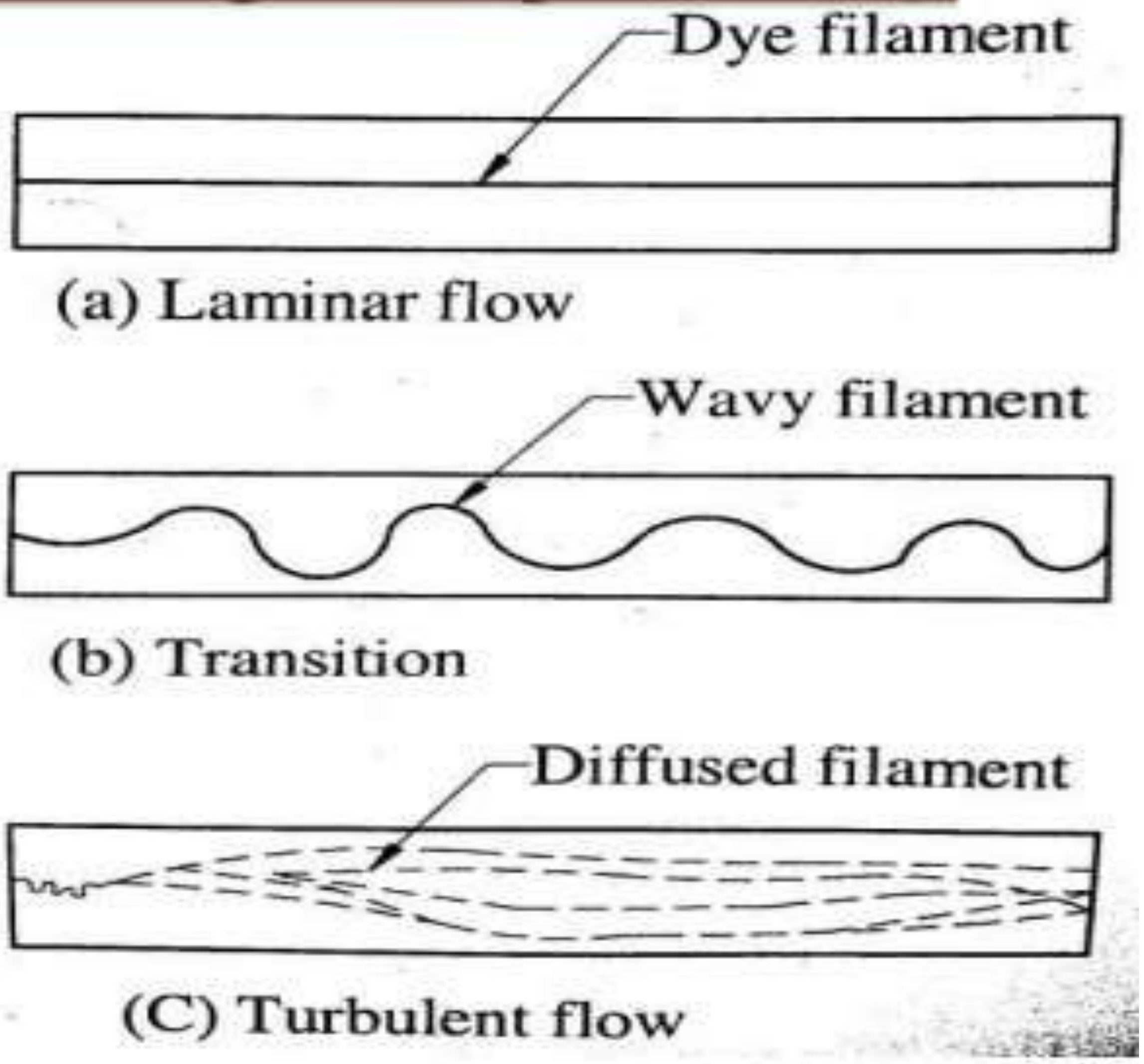
- Flow in river at the time of flood.
- Flow through pipe of different cross- section.

Observation by Reynolds

1. At low velocity, the dye will move in a line parallel to the tube and also it does not get dispersed.

2. At velocity little more than before the dye moves in a wave form.

3. At more velocity the dye will no longer move in a straight-



(a) Laminar flow
(b) Transition
(C) Turbulent flow

What is Reynolds Number ?

- The ratio of inertia force to viscous force is said to be the Reynolds number (R_N).

FORMULAS

$$R_N = \frac{\text{Inertia Force}}{\text{Viscous Force}}$$

$$R_N = \frac{\rho x V x D}{\mu}$$

$$R_N = \frac{V x D}{\nu}$$

Where

$$\nu = \frac{\mu}{\rho}$$

Where,

ρ =density of liquid (Kg/m^3)
V=mean velocity of liquid m/S
D=diameter of pipe(m)
μ=dynamic velocity($N.S/m^2$)
ν =kinematic viscosity(m^2 /S)

- **Reynold number is a dimensionless quantity**.

- If Reynold number, $R_N < 2000$ the flow is **laminar flow**.

- If Reynold number, $R_N > 4000$ the flow is **turbulent flow.**

- If Reynold number i.e. $2000 < R_N < 4000$,we observe a flow in which we can see both laminar and turbulent flow to gather. This flow is called **Transition flow.**

- $R_N = 2300$ is usually accepted as the value at transition , R_N that exists anywhere in the transition region is called the **critical Reynolds number.**

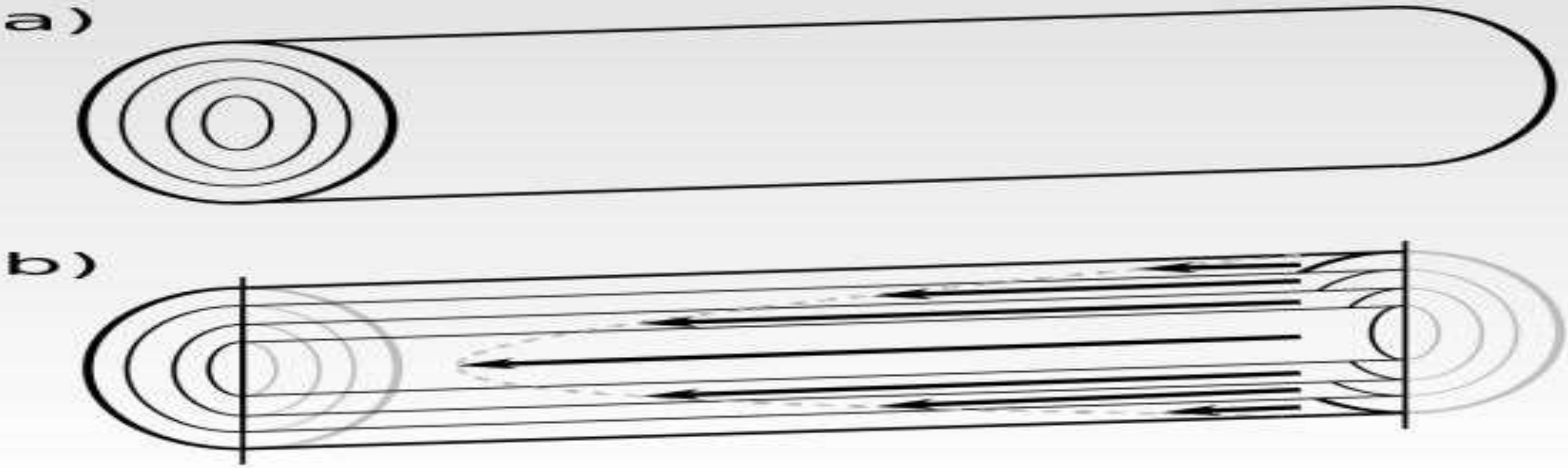
Critical Reynold’s number

- The Reynolds number at which the flow becomes turbulent is called the critical Reynolds number
- The value of the critical Reynolds number is different for different geometries and flow conditions
- For internal flow in a circular pipe , the generally accepted value of the critical Reynolds number is 2300

Laminar flow through circular pipes – *HAGEN –POISEUILLE EQUATION*

HAGEN POISEVILLE LAW

- In non ideal fluid dynamics, the Hagen–Poiseville equation, also known as the Hagen–Poiseville law, Poiseville law or Poiseville equation, is a physical law that gives the pressure drop in an incompressible and Newtonian fluid in laminar flow flowing through a long cylindrical pipe of constant cross section.



HAGEN POISEVILLE LAW

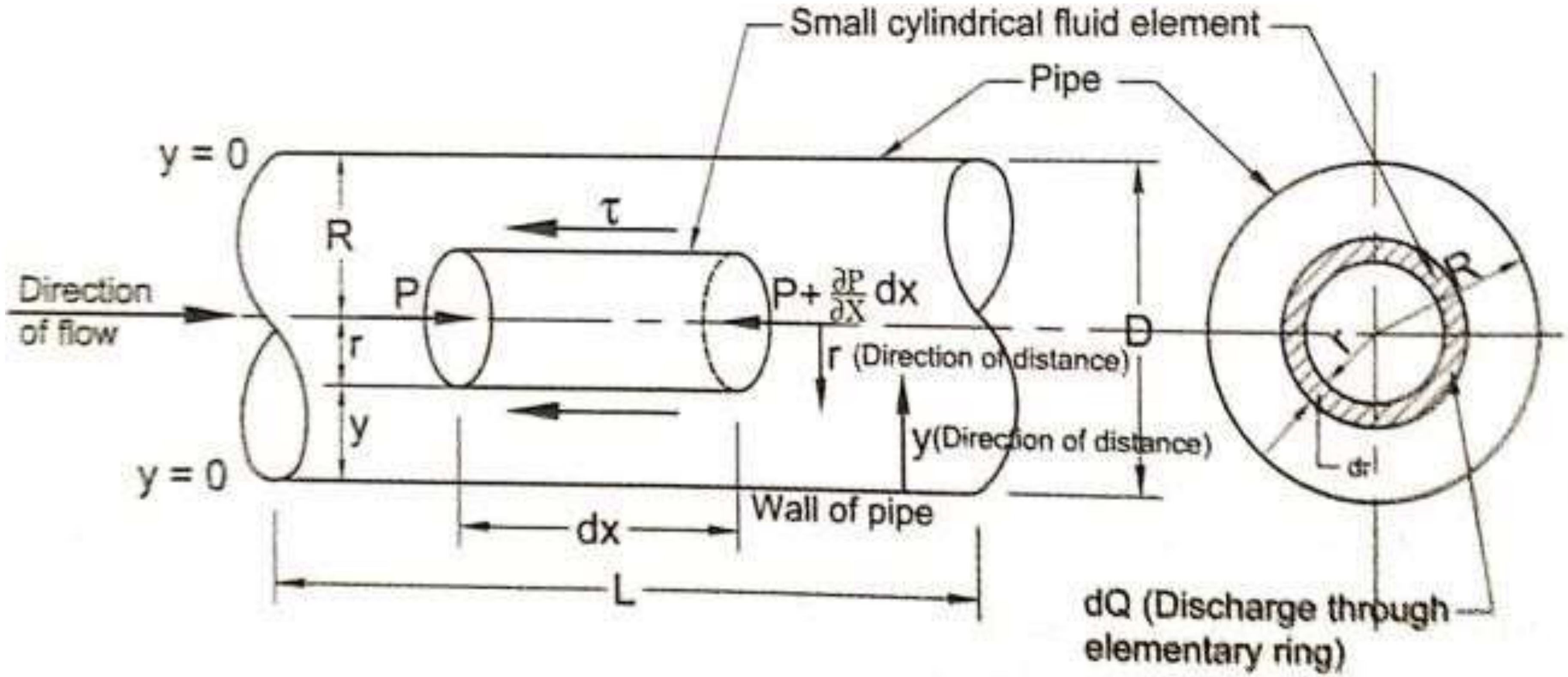
- *For analyzing laminar or viscous flow of an incompressible fluid through a circular pipe of radius R. We consider the equilibrium of forces on a small concentric cylindrical fluid element of radius r and length dx. If 'p' is the intensity of pressure on the face AB , then the intensity of pressure on face CD will be* $(p + \frac{\partial p}{\partial x}\ \partial x)$.

- *Then the forces acting on the fluid element are :*

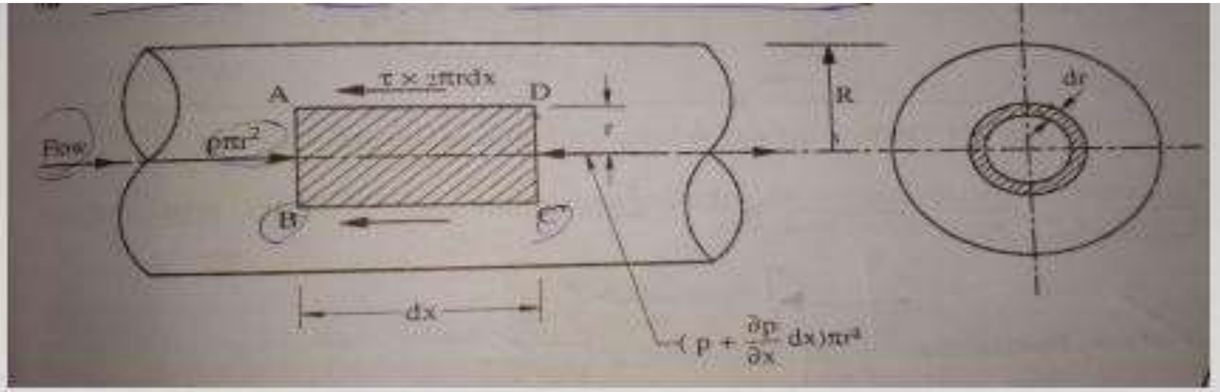
1. *The pressure force,* $p * \Pi r^2$ *on face AB.*

2. *The pressure force ,* $(p + \frac{\partial p}{\partial x}\ \partial x)\Pi r^2$ *on face CD.*

3. *The shear force,* $\tau * 2\Pi r\, dx$ *on the surface of fluid element.*



Small cylindrical fluid element
Pipe
y = 0
τ
R
Direction
of flow
P
P+ $\frac{\partial p}{\partial x}$ dx
r
r (Direction of distance)
D
y
y(Direction of distance)
y = 0
Wall of pipe
dx
L
R
r
dr
dQ (Discharge through
elementary ring)



- *As there is no acceleration, hence the summation of all the forces in the direction of flow must be zero, i.e.*

$$p * \Pi r^2 - (p + \frac{\partial p}{\partial x} \partial x)\Pi r^2 - \tau * 2\Pi r\, dx = 0$$

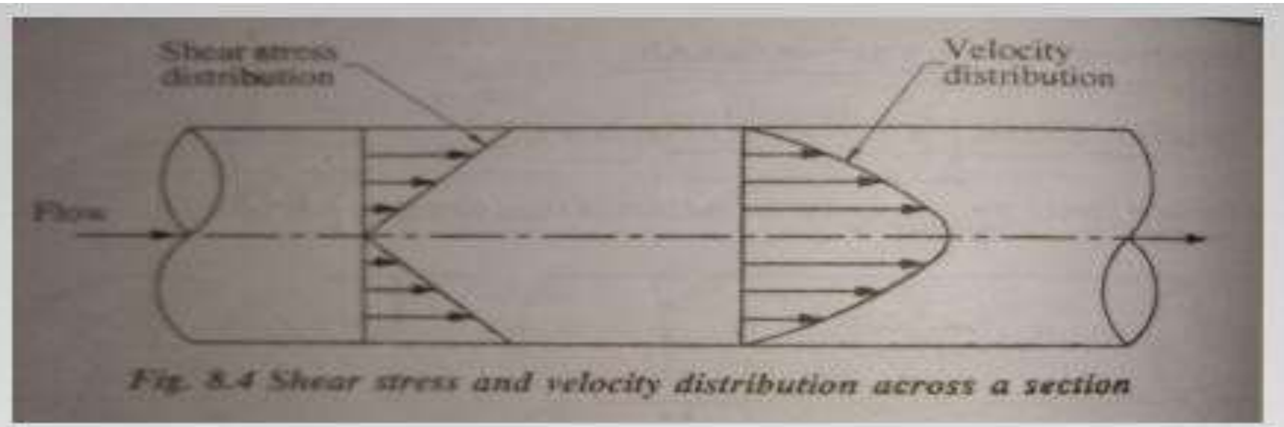
or

$$-\frac{\partial p}{\partial x} \partial x \Pi r^2 - \tau * 2\Pi r\, dx = 0$$

$$-\frac{\partial p}{\partial x} r - 2\tau = 0$$

$$\tau = -\frac{\partial p}{\partial x}\frac{r}{2} \text{ ....}(1)$$

- *The shear stress τ across a section varies with 'r' as $\frac{\partial p}{\partial x}$ across a section is constant. Hence it is linear.*



Equation (1) indicates that the value of shear stress is zero at the center of pipe (r = 0) and maximum at the pipe wall (r = R) given by

$$\tau_{max} = -\frac{\partial p}{\partial x}\left(\frac{R}{2}\right) \ldots\ldots\ldots (2)$$

(a) <u>Velocity Distribution:</u>

To obtain the velocity distribution across a section, the value of shear stress $\tau = \mu \frac{du}{dy}$ *is substituted in equation (1).*

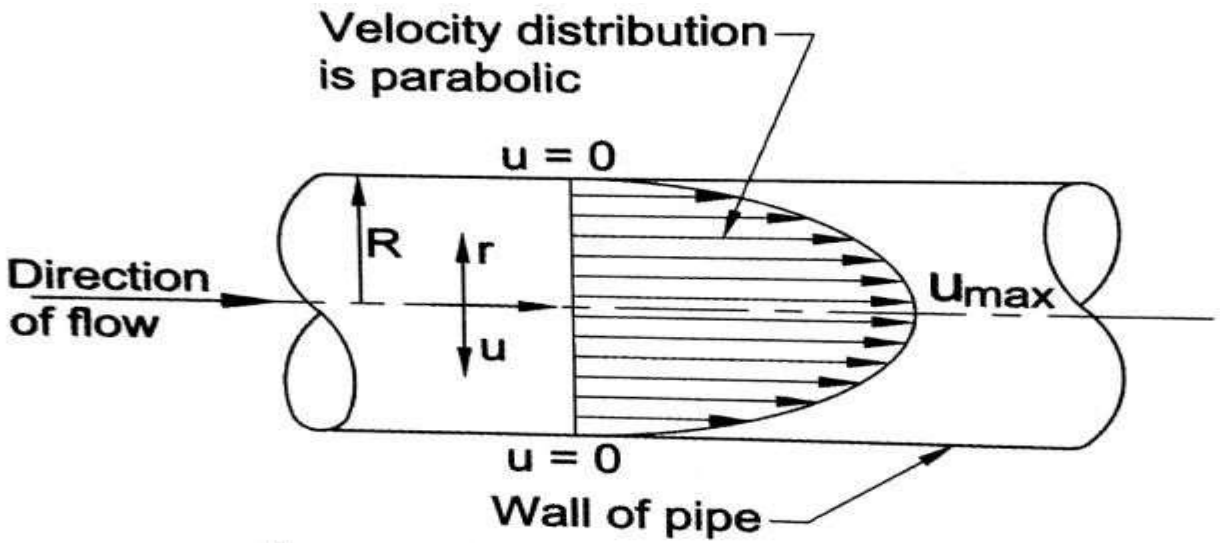


Fig. Velocity distribution for laminar flow through pipe

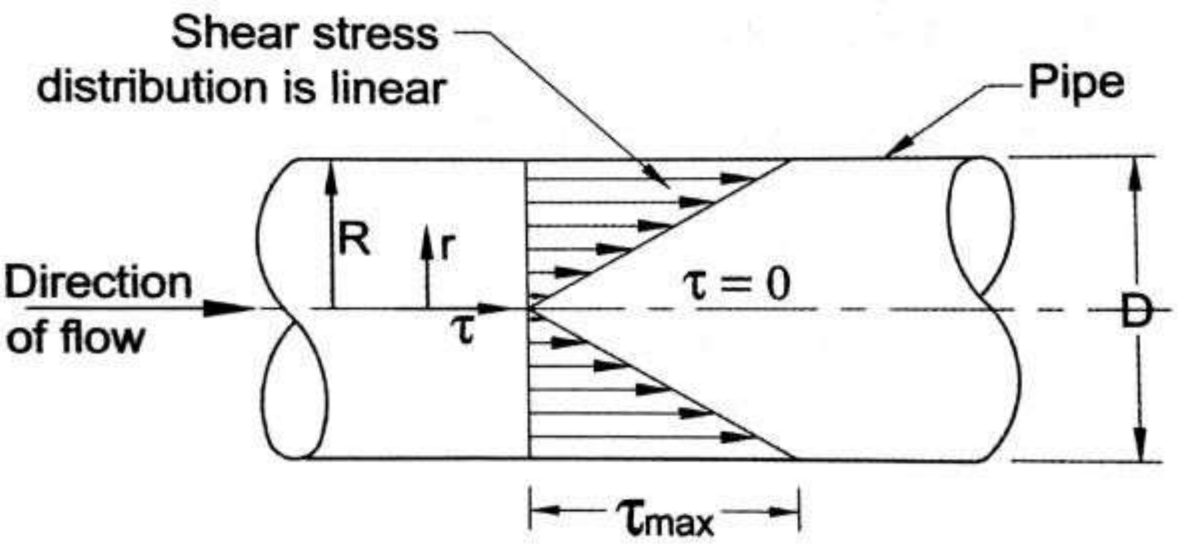


Fig. Shear stress distribution for laminar flow through pipe

- But the relation $\tau = \mu \frac{du}{dy}$, y is measured from the pipe wall.

$$y = R - r \qquad and \quad dy = - dr$$

$$\tau = \mu \frac{du}{-dr} = - \mu \frac{du}{dr}$$

The minus sign indicates r is measured in opposite of y, substituting the value in equation (1) we get.

$$- \mu \frac{du}{dr} = - \frac{\partial p}{\partial x} \frac{r}{2} \qquad or \qquad \frac{du}{dr} = \frac{1}{2\mu} \frac{\partial p}{\partial x} r$$

Integrating this equation w.r.t., 'r' , we get

$$u = \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 + c \text{ ....(3)}$$

where c is the constant of integration and its value is obtained from the boundary condition that at $r = R$, $u = 0$

$$0 = \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 + c$$

$$c = - \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2$$

Substituting the value of c in equation (3) ,we get

$$u = \frac{1}{4\mu}\frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$$

$$u = -\frac{1}{4\mu}\frac{\partial p}{\partial x}[R^2 - r^2]\text{..........}(4)$$

The values of μ, $\frac{\partial p}{\partial x}$ *and R are constants in above equation, which means the velocity it varies with square of r.*

(b) <u>Ratio of maximum velocity to average velocity:</u>

From the equation 4 , the velocity is maximum when r = 0.Thus maximum velocity Umax is obtained as

$$U_{max} = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2\text{.............}(5)$$

The fluid flowing per second through the elementary ring,

*dQ = velocity at a radius r * area of the ring element*

$$= u * 2\Pi r\, dr$$

$$= - \frac{1}{4\mu}\frac{\partial p}{\partial x}[R^2 - r^2] * 2\Pi r\, dr$$

$$Q = \int_0^R dQ = \int_0^R - \frac{1}{4\mu}\ \frac{\partial p}{\partial x}[R^2 - r^2] * 2\Pi r\, dr$$

$$= - \frac{1}{4\mu}\frac{\partial p}{\partial x} * 2\Pi \int_0^R [R^2 - r^2]\, r dr$$

$$= \frac{1}{4\mu}(- \frac{\partial p}{\partial x}) * 2\Pi \int_0^R [R^2 r - r^3]\, dr$$

$$= \frac{1}{4\mu}(- \frac{\partial p}{\partial x}) * 2\Pi[\frac{R^2 r^2}{2} - \frac{r^4}{4}]$$

$$= \frac{1}{4\mu}(- \frac{\partial p}{\partial x}) * 2\Pi[\frac{R^4}{2} - \frac{R^4}{4}]$$

$$= \frac{1}{4\mu}(- \frac{\partial p}{\partial x}) * 2\Pi * \frac{R^4}{4} = \frac{\frac{\Pi}{8\mu}(- \frac{\partial p}{\partial x})R^4}{\pi R^2}$$

$$\bar{u} = \frac{1}{8\mu}(- \frac{\partial p}{\partial x})R^2 \text{.............}(6)$$

Dividing of equations by (5) by equation (6)

Ratio of maximum velocity to average velocity = 2

(c) Drop of pressure for a given length (L) of a pipe:

From equation (6), we have

$$\bar{u} = \frac{1}{8\mu}\left(-\frac{\partial p}{\partial x}\right)R^2 \quad or \quad \left(-\frac{\partial p}{\partial x}\right) = \frac{8\mu\bar{u}}{R^2}$$

Integrating the above equation w.r.t. 'x' , we get

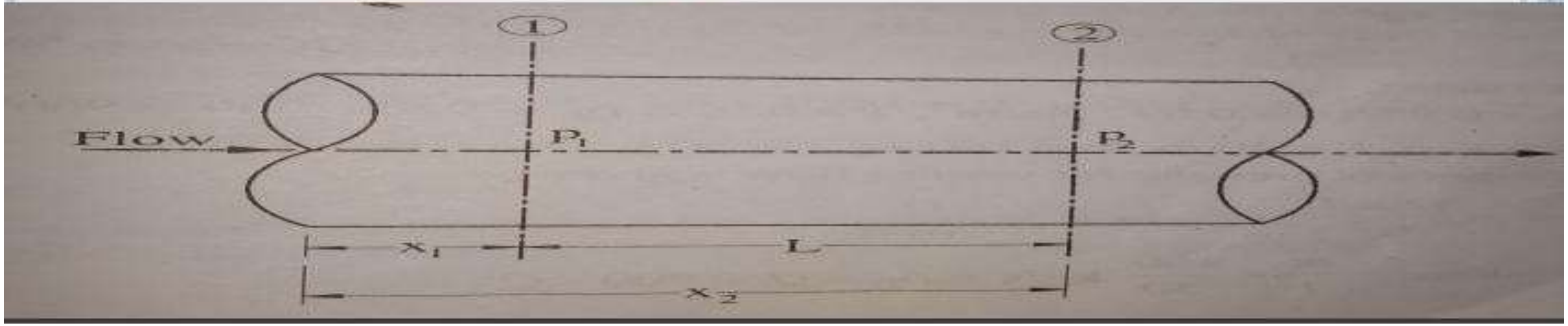
$$-\int_1^2 dp = \int_1^2 \frac{8\mu\bar{u}}{R^2}dx$$

$$-[p_1 - p_2] = \frac{8\mu\bar{u}}{R^2}[x_1 - x_2]$$

Or $$[p_1 - p_2] = \frac{8\mu\bar{u}}{R^2}[x_2 - x_1]$$

$$= \frac{8\mu\bar{u}}{R^2} L$$

$$= \frac{8\mu\bar{u}}{\left(\frac{D}{2}\right)^2} L$$



or $(p_1 - p_2) = \frac{32\mu\bar{u}}{(D)^2}L$, where $p_1 - p_2$ is the drop of pressure.

Loss of pressure head $= \frac{p_1 - p_2}{\rho g}$

$$\boxed{\frac{p_1 - p_2}{\rho g} = hf = \frac{32\mu\bar{u}}{\rho g D^2}L}$$

This equation is known as HAGEN – POISEVILLE LAW.

RELATION BETWEEN 'f' AND 'Re'

$$h_f = \frac{32\mu VL}{\rho g D^2} \quad \text{...(1)}$$

$$h_f = \frac{4fLV^2}{2gD} \quad \text{...(2)}$$

$$\frac{4fLV^2}{2gD} = \frac{32\mu VL}{\rho g D^2}$$

$$4f = \frac{32\mu VL}{\rho g D^2} \times \frac{2gD}{LV^2}$$

$$4f = \frac{64\mu}{\rho VD} = \frac{64}{\left(\frac{\rho VD}{\mu}\right)}$$

We know that, the term $\left(\frac{\rho VD}{\mu}\right)$ = Reynold's number (Re)

$$4f = \frac{64}{Re} \quad \text{for Re < 2000 (Laminar flow)}$$

where, 4f = Friction factor

$$f = \frac{16}{Re} \quad \text{for Re < 2000 (Laminar flow)}$$

where, f = Darcy's – friction co-efficient

Re = Reynold's number

Example 1: Oil of absolute viscosity 1.5 poise and relative density 0.85 flows through a 30 cm diameter pipe. If the head loss in 3000 m length of pipe is 20 m. Estimate

(a) The shear stress at the pipe wall.

(b) Shear stress at a radial distance of 10 cm from the pipe axis.

(c) The friction factor (4f) by assuming the flow to be laminar

Given data:

Absolute viscosity of oil, $\mu = 1.5$ poise $= \dfrac{1.5}{10} = 0.15$ N-s/m^2

Relative density of oil, $S = 0.85$

Diameter of the pipe, $D = 30$ cm $= 0.3$ m

Length of the pipe, $L = 3000$ m

Head loss due to friction, $h_f = 20$ m

Solution:

The equation for shear stress distribution is given by. $\tau = \left(-\frac{\partial P}{\partial x}\right)\left(\frac{r}{2}\right)$

$-\partial P$ = Pressure drop in the direction of flow in $N/m^2 = (P_1 - P_2)$

Pressure drop in metres of oil (head loss due to friction) = h_f

$$h_f = \frac{(P_1 - P_2)}{\rho g}$$

$(P_1 - P_2) = h_f \times \rho g$

ρ = Density of oil = Specific gravity of oil × density of water

$\rho = 0.85 \times 1000 = 850 \text{ kg/m}^3$

$(P_1 - P_2) = 20 \times 850 \times 9.81 = 166770 \text{ N/m}^2$

$\partial x = (x_2 - x_1)$ = Length of the pipe = 3000 m

$\left(-\frac{\partial P}{\partial x}\right)$ = Pressure drop per unit length of the pipe $= \frac{16670}{3000} = 55.59$

· = Radial distance from the centre of the pipe.

At the wall of the pipe, r = R (Radius of pipe)

Shear stress at the wall of the pipe is given by, $\tau_{max:} = \left(-\frac{\partial P}{\partial x}\right)\left(\frac{R}{2}\right)$

R = Radius of the pipe $= \left(\frac{D}{2}\right) = \left(\frac{0.3}{2}\right) = 0.15$ m

$$\tau_{max:} = 55.59 \times \frac{0.15}{2} = 4.169 \text{ N/m}^2$$

Shear stress at a radial distance of 10 cm from the pipe axis. r = 10 cm = 0.1 m

$$\tau = \left(-\frac{\partial P}{\partial x}\right)\left(\frac{r}{2}\right) = 55.59 \times \frac{0.1}{2} = 2.779 \text{ N/m}^2$$

According to Hagen-Poiseuillie equation, the head loss due to friction is given by

$$h_f = \frac{32 \mu V L}{\rho g D^2}$$

$$20 = \frac{32 \times 0.15 \times V \times 3000}{800 \times 9.81 \times (0.3)^2} = 19.188\ V$$

$$V = \frac{20}{19.188} = 1.042\ \text{m/s}$$

Average velocity of oil flowing through the pipe, $V = 1.042$ m/s

Reynold's number of flow, $Re = \dfrac{\rho V D}{\mu} = \dfrac{850 \times 1.042 \times 0.3}{0.15} = 1771.4$

$$4f = \frac{64}{Re} = \frac{64}{1771.4} = 0.0361$$

2.

HW

Glycerine flows at a velocity of 5 m/s in a 10 cm diameter pipe.

Estimate:-

(a) The boundary shear stress in the pipe due to the flow.

(b) Head loss in a length of 12 m of pipe.

(c) Power expended by the flow in a distance of 12 m.

Assume dynamic viscosity of glycerine as 1.50 Pa.s and density of glycerine as 1260 kg/m^3

DARCY- WEISBACH EQUATION

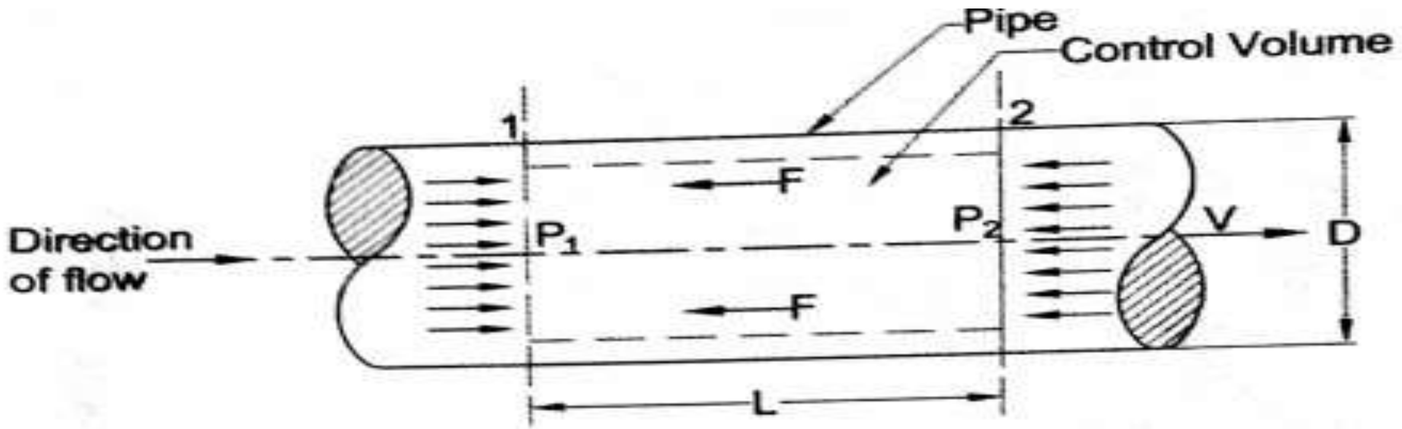


Fig.12.4 Loss of head in a pipe

Figure 12.4 shows a horizontal pipe having steady flow. Consider a control volume enclosed between sections (1) and (2).

Let P_1 = Intensity of pressure at section 1

P_2 = Intensity of pressure at section 2

V_1 = Velocity of flow at section 1

V_2 = Velocity of flow at section 2

L = Length of the pipe between sections 1 and 2

D = Diameter of the pipe

h_f = Head loss due to friction

By applying Bernoulli's equation between the sections 1 and 2, we obtain

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2 + h_f$$

$Z_1 = Z_2$ (For horizontal pipe)

$V_1 = V_2$ (For uniform diameter of pipe, no change in velocity)

Therefore, $\frac{P_1}{\rho g} = \frac{P_2}{\rho g} + h_f$

Loss of head due to friction, $h_f = \frac{P_1}{\rho g} - \frac{P_2}{\rho g}$

$$h_f = \frac{P_1 - P_2}{\rho g} \quad \text{..(1)}$$

Therefore, Total frictional resistance, $F = f^1 \times \pi D L \times V^n$

For commercial pipes, $n = 2$

$$F = f^1 \times \pi D L \times V^2 \quad \text{..(2)}$$

The various forces acting on the fluid between sections (1) and (2).

(i) Pressure force at section 1 $= P_1 \times A_1$

(ii) Pressure force at section 2 $= P_2 \times A_2$

(iii) Frictional resistance, $F = f^1 \times \pi D L \times V^2$

Resolving all the forces in the direction of flow we have,

$P_1 A_1 - P_2 A_2 - F = 0$

$A_1 = A_2 = A$ (For uniform diameter pipe). Then $P_1 A - P_2 A = F$

$(P_1 - P_2) A = F$

$(P_1 - P_2) A = f^1 \times \pi D L \times V^2$

From equation (1), $h_f = \frac{P_1 - P_2}{\rho g}$

$$(P_1 - P_2) = h_f \times \rho g$$

Therefore, $h_f \times \rho g \times A = f^1 \times \pi D L \times V^2$

$$h_f = \frac{f^1}{\rho g} \times \frac{\pi D L}{A} \times V^2 \quad \text{..(3)}$$

But, A = Cross-sectional area of pipe = $\frac{\pi}{4} D^2$

$$h_f = \frac{f^1}{\rho g} \times \frac{\pi D L \times 4}{\pi D^2} \times V^2 = \frac{f^1}{\rho g} \times \frac{4L}{D} \times V^2$$

Multiplying the numerator and denominator by (2g)

$$h_f = \left(\frac{f^1 \times 2g}{\rho g} \right) \times \frac{4L}{D} \times \frac{V^2}{2g}$$

The term $\left(\frac{2g f^1}{\rho g} \right)$ is a non-dimensional quantity.

It is called Darcy's friction co-efficient 'f'. $f = \left(\frac{2g f^1}{\rho g} \right)$

Therefore $$\mathbf{h_f = \frac{4 f L V^2}{2 g D}}$$

This equation is known as Darcy-Weisbach equation.

Friction factor and Darcy's friction co-efficient are function of Reynold's number.

For laminar flow:- (Re < 2000)

Darcy's friction co-efficient, $f = \frac{16}{Re}$ Friction factor, $4f = \frac{64}{Re}$

For turbulent flow:- (Re > 4000)

Darcy's friction co-efficient, $f = \frac{0.079}{(Re)^{0.25}}$ Friction factor, $4f = \frac{0.316}{(Re)^{0.25}}$

1 An oil of specific gravity 0.85 and viscosity 0.05 poise flows through a 20 cm diameter pipe at the rate of 75 litre per second. Find the head lost due to friction for a 500 m length of pipe. Also calculate the power required to maintain the flows.

Given data:

Specific gravity of oil = 0.85

Viscosity of oil, $\mu = 0.05$ poise $= \frac{0.05}{10} = 0.005\ N\text{-}s/m^2$

Diameter of pipe, $D = 20\ cm = 0.2\ m$

Rate of flow of oil, $Q = 75$ litre/s $= 0.075\ m^3/s$

Reynold's number is given by, $Re = \dfrac{\rho V D}{\mu}$

ρ = Density of oil = Specific gravity of oil × Density of water

$= 0.85 \times 1000 = 850 \text{ kg/m}^3$

V = Velocity of flow $= \dfrac{Q}{A}$

where, A = Cross-sectional area of pipe $= \dfrac{\pi}{4}D^2 = \dfrac{\pi}{4}(0.2)^2 = 0.0314 \text{ m}^2$

$$V = \frac{0.075}{0.0314} = 2.388 \text{ m/s}$$

$$Re = \frac{850 \times 2.388 \times 0.2}{0.005} = 81210$$

$$\text{Fiction factor } (4f) = \frac{0.316}{(Re)^{0.25}} = \frac{0.316}{(81210)^{0.25}} = 0.01871$$

Head lost due to friction can be calculated by using Darcy-Weisbach equation.

$$h_f = \frac{4 f L V^2}{2 g D}$$

$$h_f = \frac{0.01871 \times 500 \times (2.388)^2}{2 \times 9.81 \times 0.2} = \underline{13.6 \text{ m of oil}}$$

Certain amount of energy possessed by the flowing fluid will be consumed in overcoming the frictional resistance to the flow. This energy loss is called power loss due to friction.

Power loss due to friction = $\rho\, g\, Q\, h_f$

where, ρ = Density of oil = Specific gravity of oil × Density of water

$= 0.85 \times 1000 = 850 \text{ kg/m}^3$

Power loss due to friction = $850 \times 9.81 \times 0.075 \times 13.6$ = 8505.27 Watts

Power required to maintain the flow = Power loss due to friction = 8505.27 Watts

2 A pipe line of 20 cm in diameter and 1500 m long carries water from one end to other. The pressure measured at inlet and outlet of the pipe line are 12 KPa and 2 KPa respectively. Determine the rate of flow of water through the pipe in litre/min. Taking Darcy's friction co-efficient for the pipe is equal to 0.008.

Given data:

Diameter of pipe, $D = 20 \text{ cm} = 0.20 \text{ m}$

Length of the pipe, $L = 1500 \text{ m}$

Pressure at inlet of the pipe, $P_1 = 12 \text{ KPa} = 12000 \text{ N/m}^2$

Pressure at outlet of the pipe, $P_2 = 2 \text{ KPa} = 2000 \text{ N/m}^2$

Darcy's friction co-efficient, $f = 0.008$

Darcy-Weisbach equation for finding head loss due to friction is given by,

$$h_f = \frac{4fLV^2}{2gD}$$

$$h_f = \text{Head loss due to friction} = \frac{(P_1 - P_2)}{\rho g}$$

where, $(P_1 - P_2)$ = Pressure drop between the inlet and outlet of the pipe

$$(P_1 - P_2) = 12000 - 2000 = 10{,}000 \text{ N/m}^2$$

$$\rho = \text{Density of water} = 1000 \text{ kg/m}^3$$

$$h_f = \frac{10000}{1000 \times 9.81} = 1.019 \text{ metres of water}$$

$$1.019 = \frac{4 \times 0.008 \times 1500 \times V^2}{2 \times 9.81 \times 0.2} = 12.232\, V^2$$

$$V^2 = \frac{1.019}{12.232} = 0.0833$$

$$V = \sqrt{0.0833} = 0.288 \text{ m/s}$$

Discharge of water, $Q = A \times V$

$$A = \text{Cross-sectional area of pipe} = \frac{\pi}{4} D^2 = \frac{\pi}{4}(0.20)^2 = 0.0314 \text{ m}^2$$

$$Q = 0.0314 \times 0.288 = 0.00904 \text{ m}^3/\text{s}$$

$$= 0.00904 \times 1000 \times 60 = 542.867 \text{ litre/min}$$

MINOR LOSSES

- The minor losses of energy are those which are caused on account of the change in the velocity of flowing fluid. The change may be there in either magnitude or direction. Minor losses are caused by certain local features or disturbance, which may cause eddy formation.
- In case of long pipes these losses are usually quite small as compared with the loss of energy due to friction and hence are called as 'minor losses'.
- It can be neglected without serious error. But in short pipes these losses may overweigh the friction loss.

The minor loss of energy includes following cases:

- Loss of head due to sudden enlargement.
- Loss of head due to sudden contraction.
- Loss of head at the entrance of pipe.
- Loss of head at the exit of pipe.
- Loss of head due to an obstruction.
- Loss of head due to bend in the pipe.
- Loss of head in various pipe fittings.

Minor losses in pipeline systems

Situation	Head Loss	Explanation
1. Sudden expansion	$h_{LE} = \frac{(V_1 - V_2)^2}{2g}$	Expansion from section 1 to 2
2. Sudden contraction	$h_{LC} = K\frac{V_2^2}{2g}$	V_2 = Velocity in contracted section $K = \left(\frac{1}{C_C} - 1\right)^2 = 0.5$ (generally)
3. Entrance to a pipe from a reservoir	$h_L = 0.5\ \frac{V^2}{2g}$	V = Velocity of flow
4. At exit of a pipe	$h_L = \frac{V^2}{2g}$	V = Velocity of flow
5. Bends, elbows and pipe fittings	$h_L = K\ \frac{V^2}{2g}$	K for bend = 0.6 K for elbow = 0.9

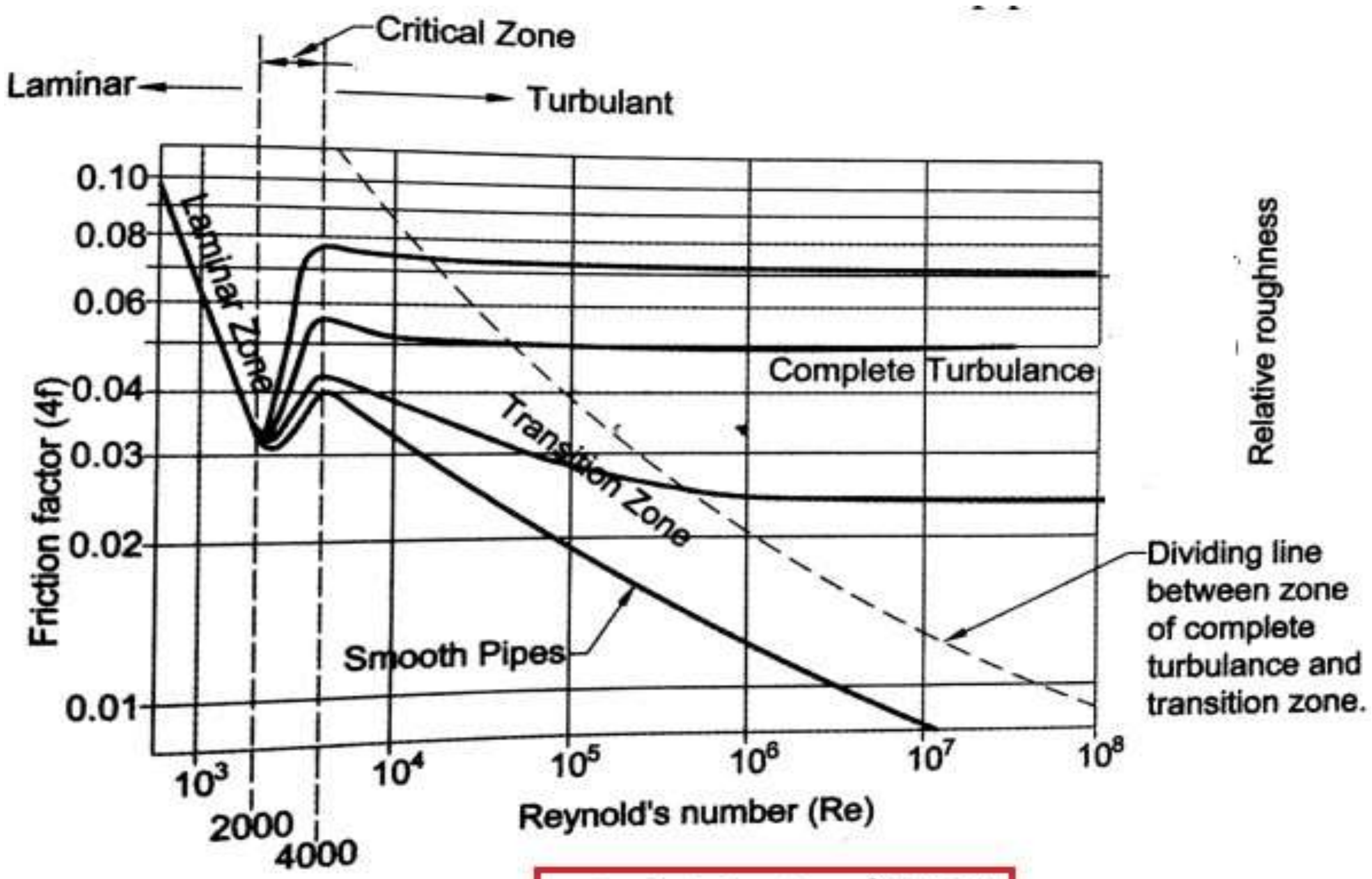
Moody's chart - *Relation between friction factor and Reynolds number*

- The chart developed by Prof. Lewis F Moody for commercial pipe has become a convenient and reliable tool for solving practical problems in pipe flow
- The Moody's chart gives the value of friction factor of any pipe provided its relative roughness and Reynolds number of flow are known

  Relative roughness = $\mathcal{E}/D$

  Where , $\mathcal{E}$ = Average roughness of pipe surface & D = diameter of pipe
- It is a graphical representation of the relation between friction factor and Reynold's number of flow for various values of relative roughness of the pipe surface

Moody's chart



Moody's chart or diagram